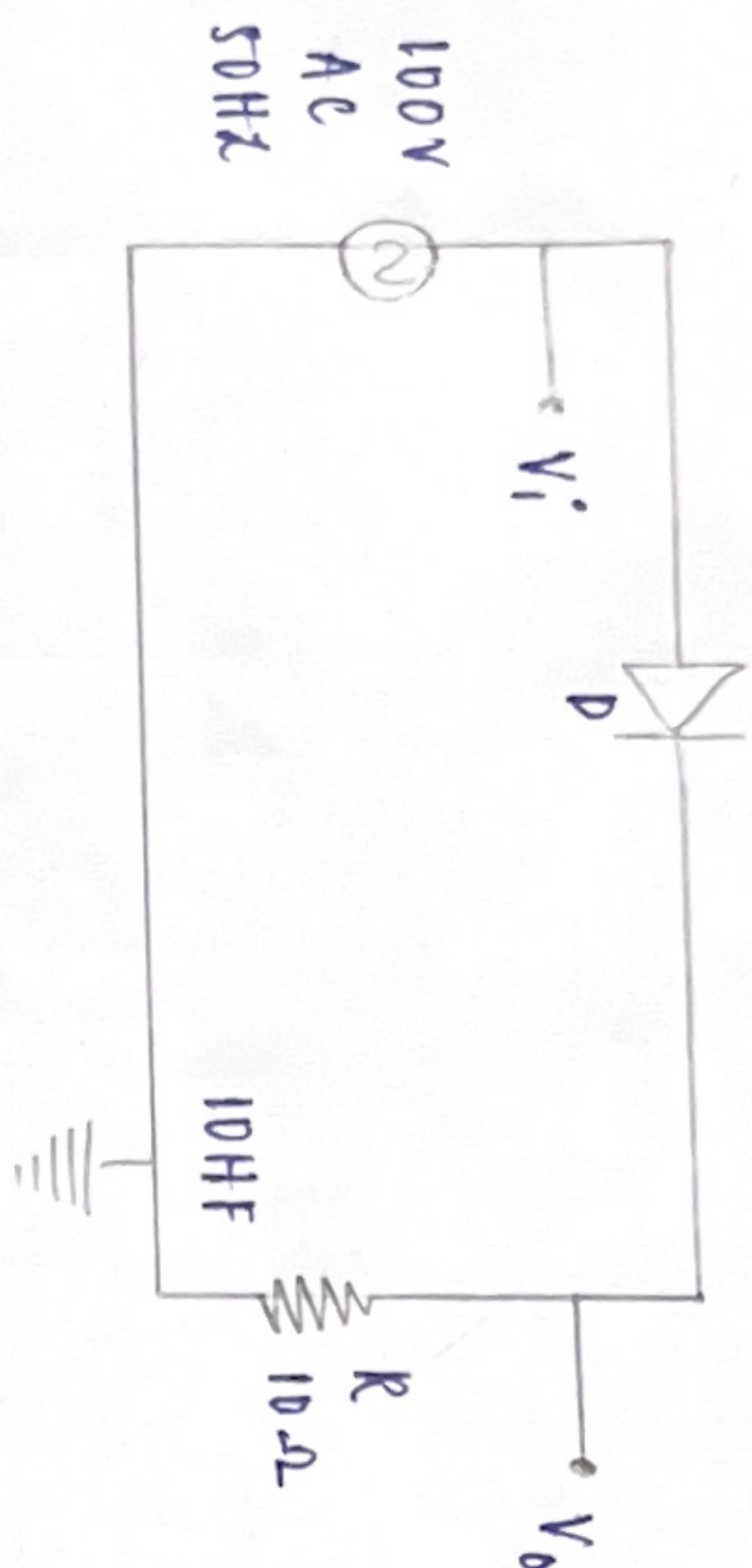
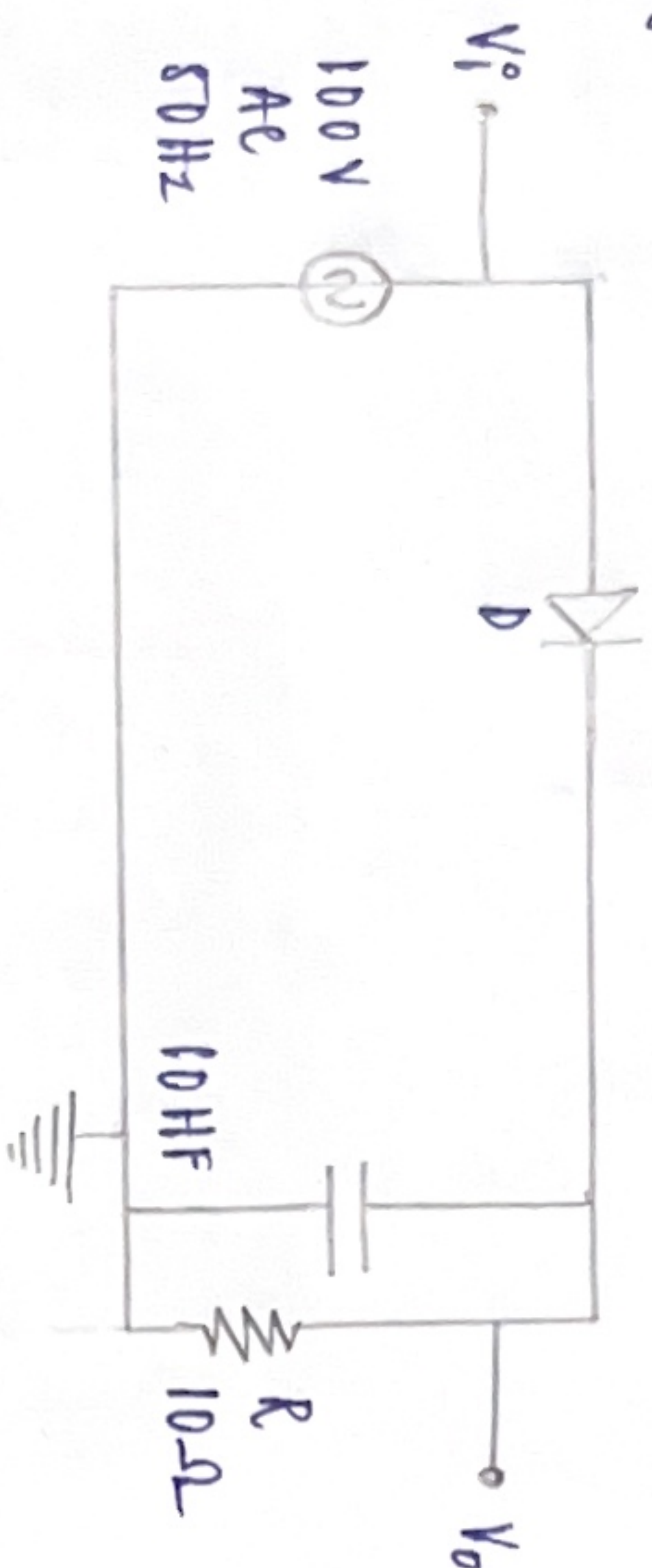


CIRCUIT DIAGRAM

HALF RECTIFIER: without filter:



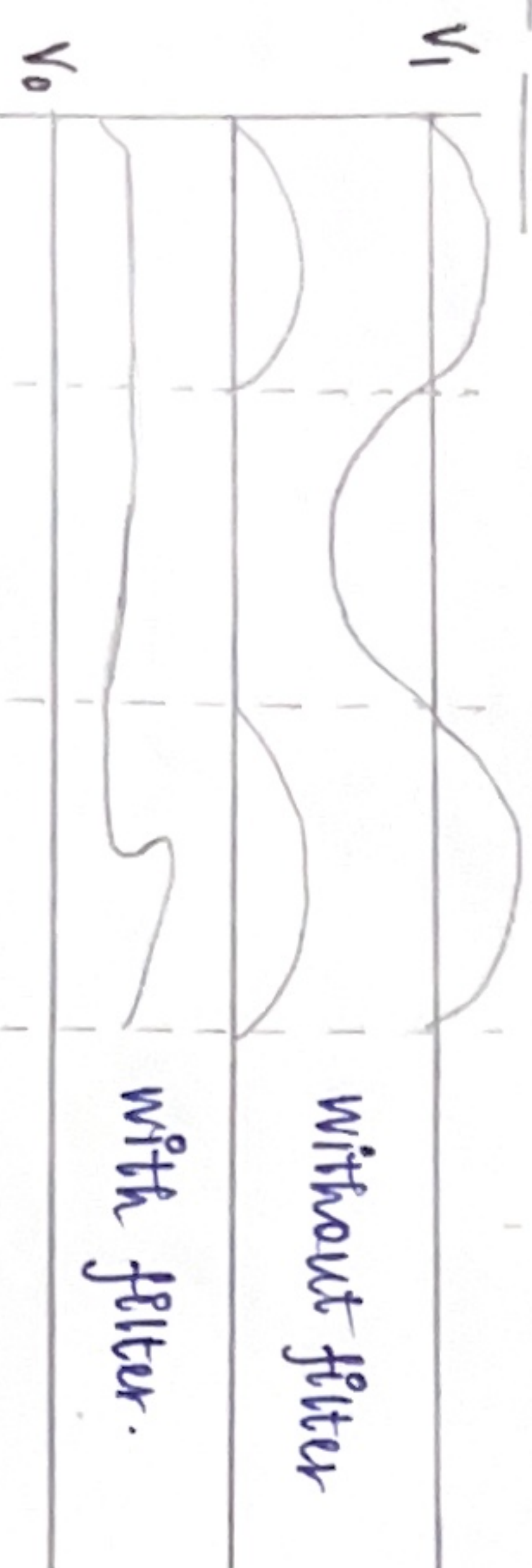
With filter:



Tabulation:

INPUT VOLTAGE	Output voltage			
	without Filter	with Filter	without Filter	with Filter
$V_m(\text{V})$	$V_m(\text{V})$	$V_m(\text{V})$	$V_m(\text{V})$	$V_m(\text{V})$
$f(\text{Hz})$	$f(\text{Hz})$	$f(\text{Hz})$	$f(\text{Hz})$	$f(\text{Hz})$
$t(\text{ms})$	$t(\text{ms})$	$t(\text{ms})$	$t(\text{ms})$	$t(\text{ms})$
100V	100V	100V	100V	85V
20ms	20ms	20ms	20ms	25ms

Model Graph:



Expt. No.

04

Page No.

26

HALF WAVE AND FULL WAVE RECTIFIER

Aim:

To simulate the following circuits in Proteus.

- (i) Half-wave rectifier
- (ii) Full-wave rectifier

Apparatus required:

Laptop with Proteus software.

Theory: A rectifier is a circuit that converts alternating current (AC) into direct current (DC).

Half-Wave Rectifier:

Half-wave rectifier conducts only during positive half cycle. During positive half cycle the diode conducts and the output voltage is equal to the input voltage. During negative half-cycle, the diode does not conduct and the output voltage is equal to zero.

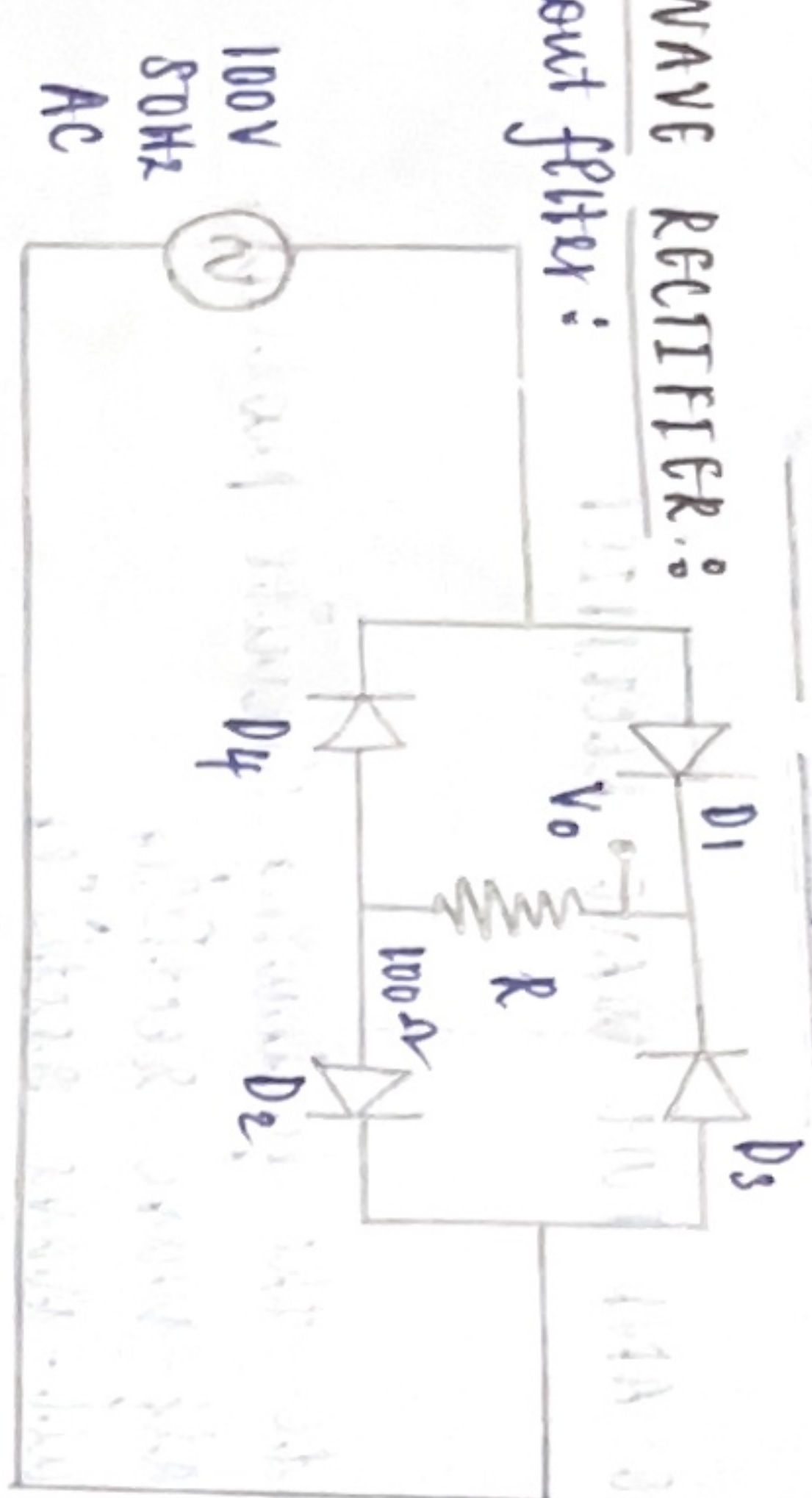
Full-Wave Rectifier:

Full wave rectifier conducts during both positive and negative half-cycles. During positive half-cycle, the output voltage is equal to input voltage during negative half-cycle the output voltage is equal to negative of the input voltage. The output of the rectifier circuit is pure D.C.

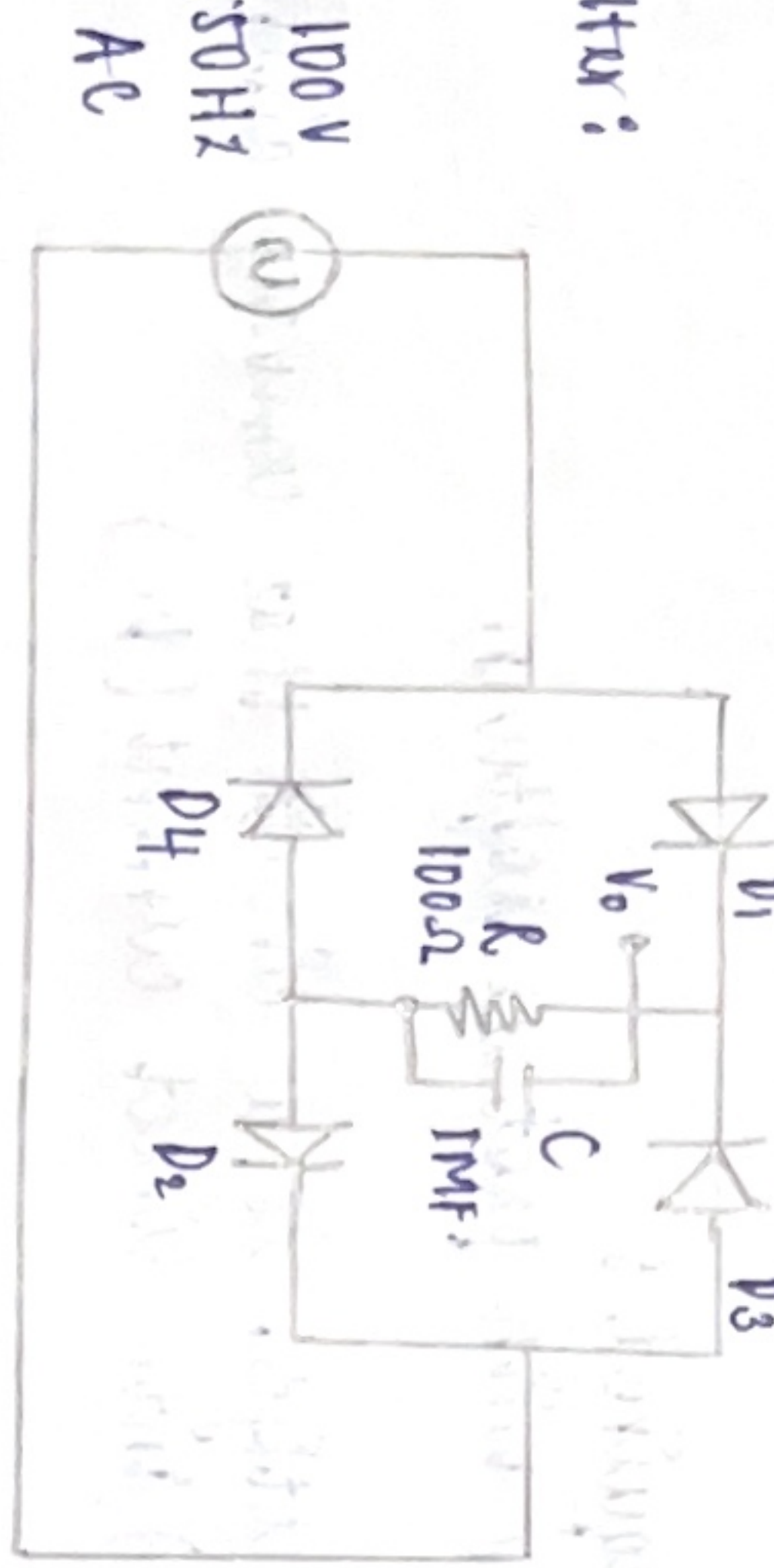
CIRCUIT DIAGRAM

FULL WAVE RECTIFIER:

Without filter:



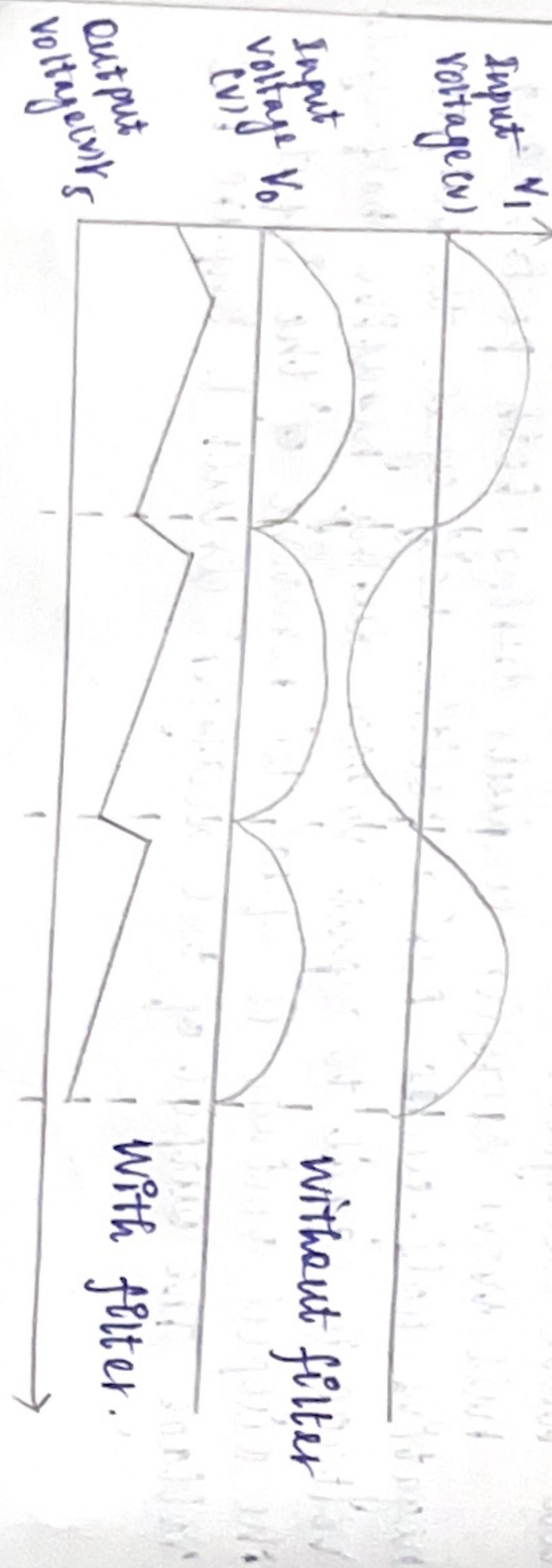
With filter:



Tabulation:

INPUT VOLTAGE		OUTPUT VOLTAGE	
		without Filter	With filter.
$V_m(V)$	$t(ms)$	$V_m(V)$	$t(ms)$
50V	20ms	50V	5ms
		100V, 46V	5ms, 24ms

Model Graph:



Expt. No.

04

Page No.

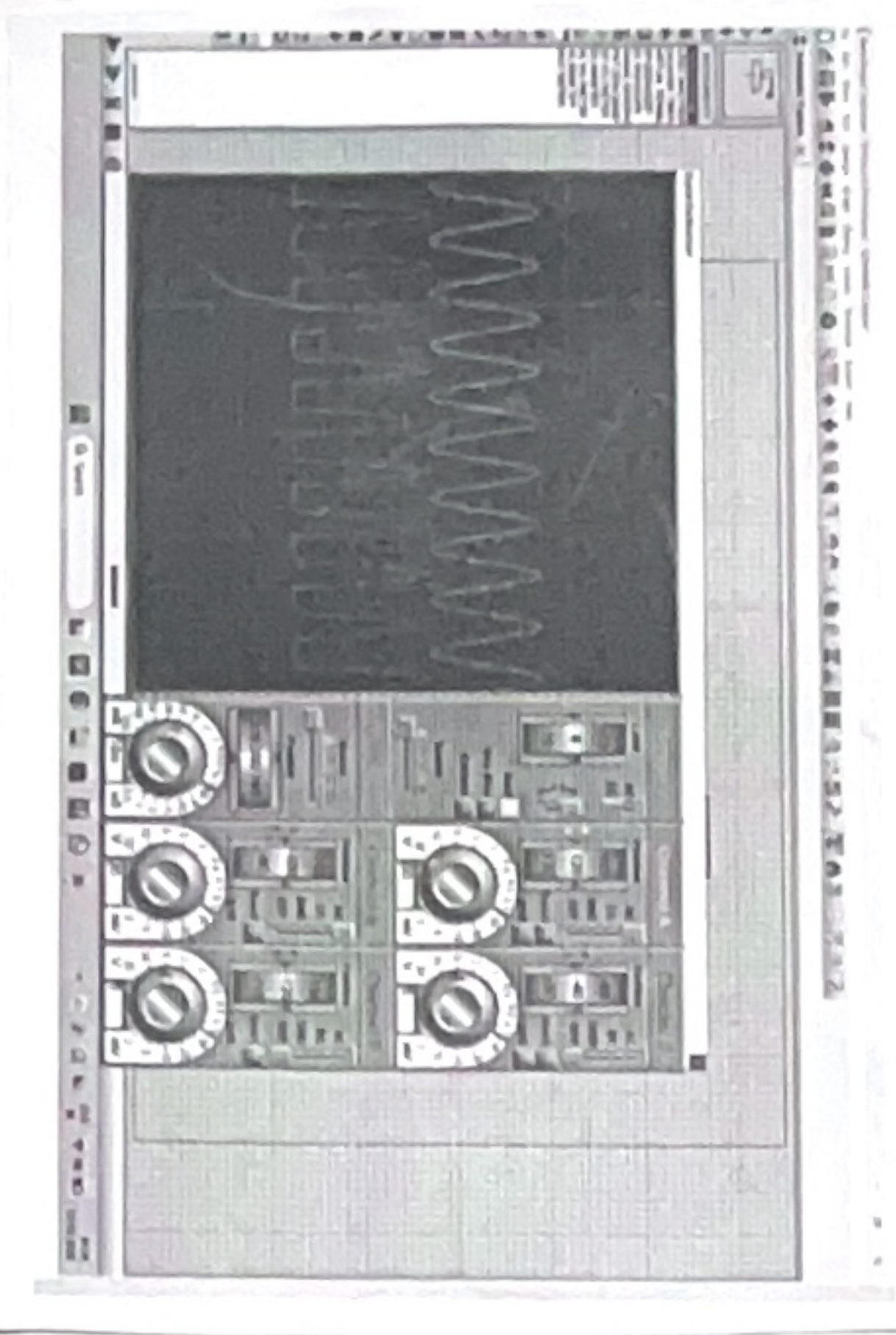
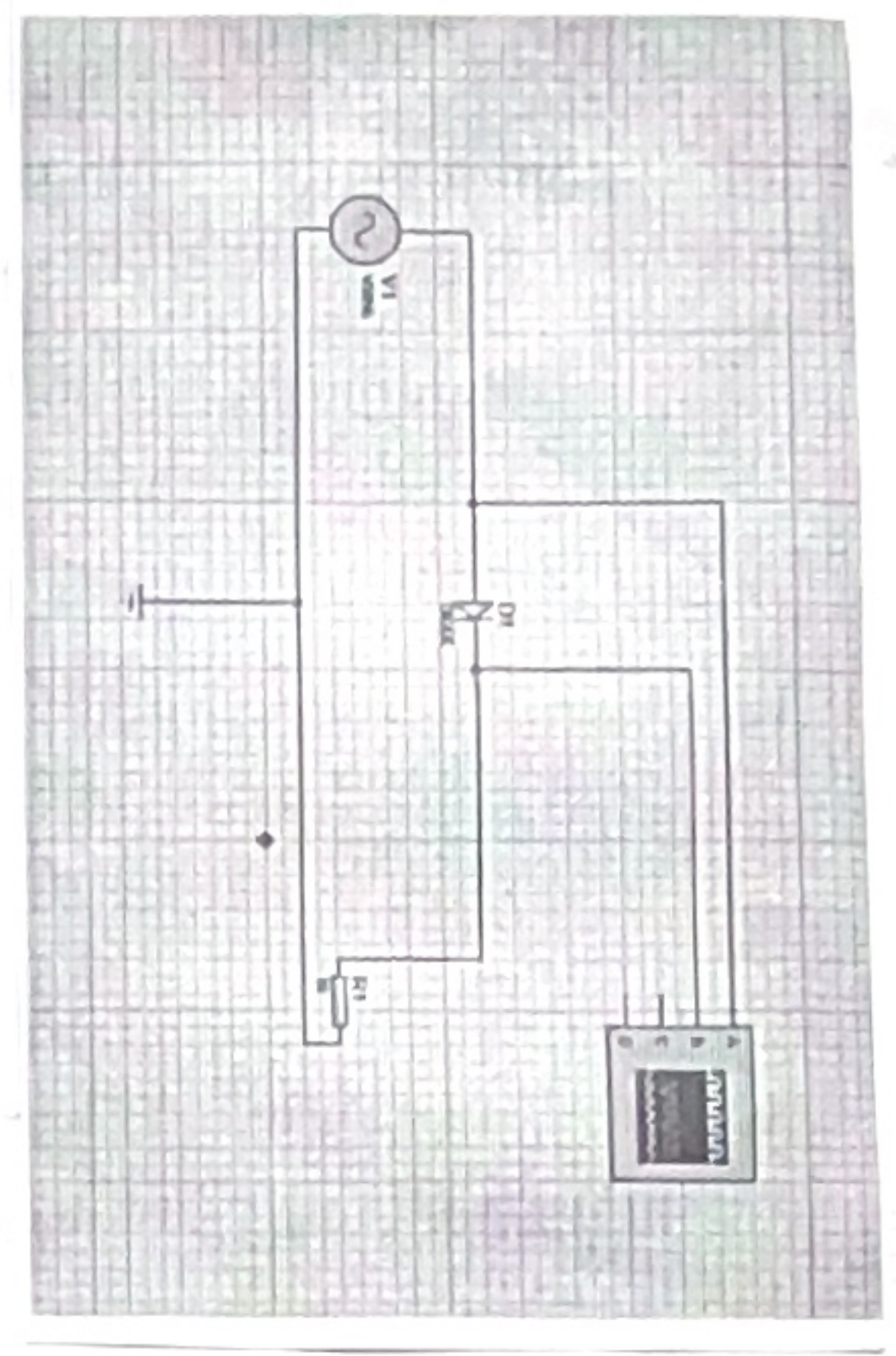
28

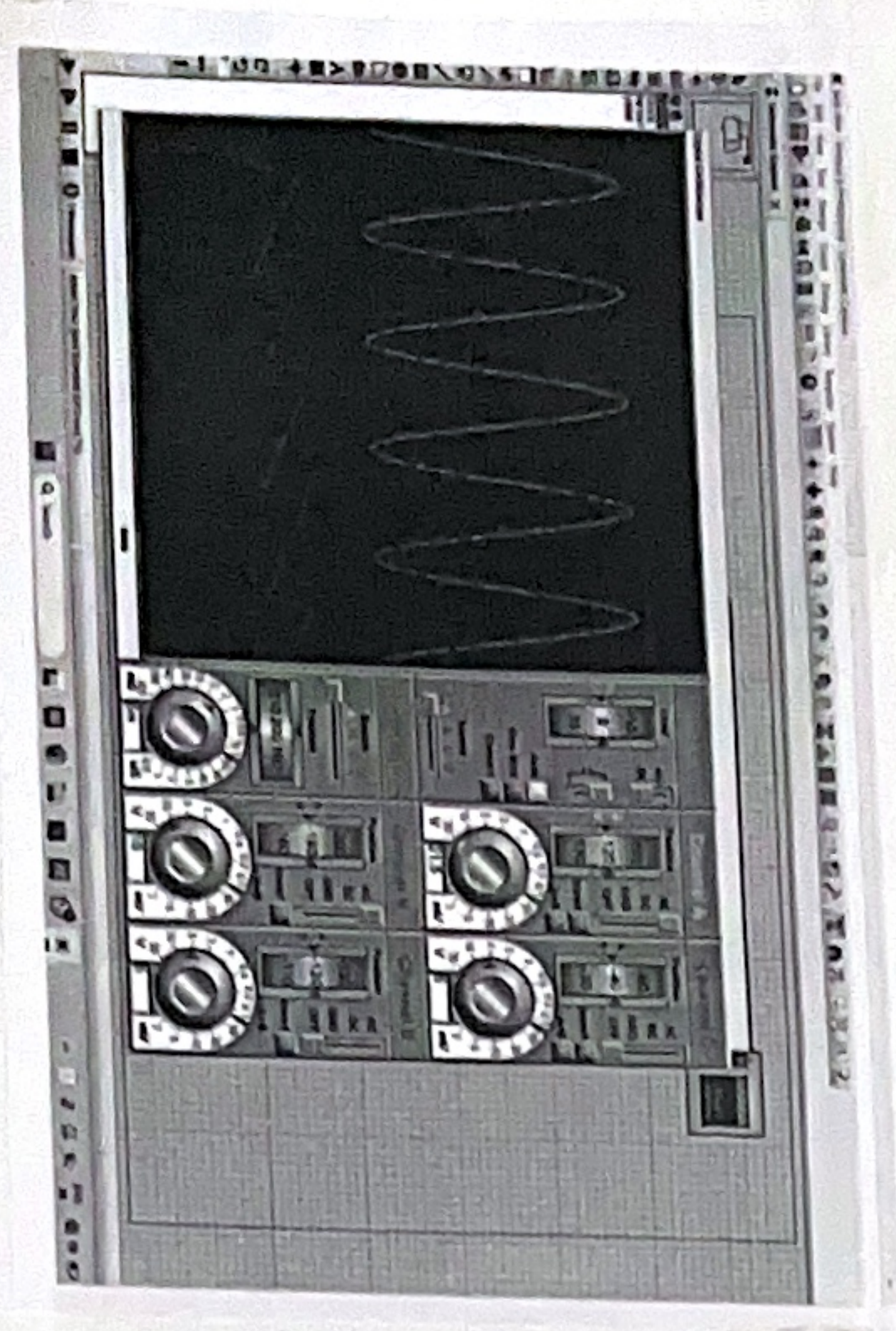
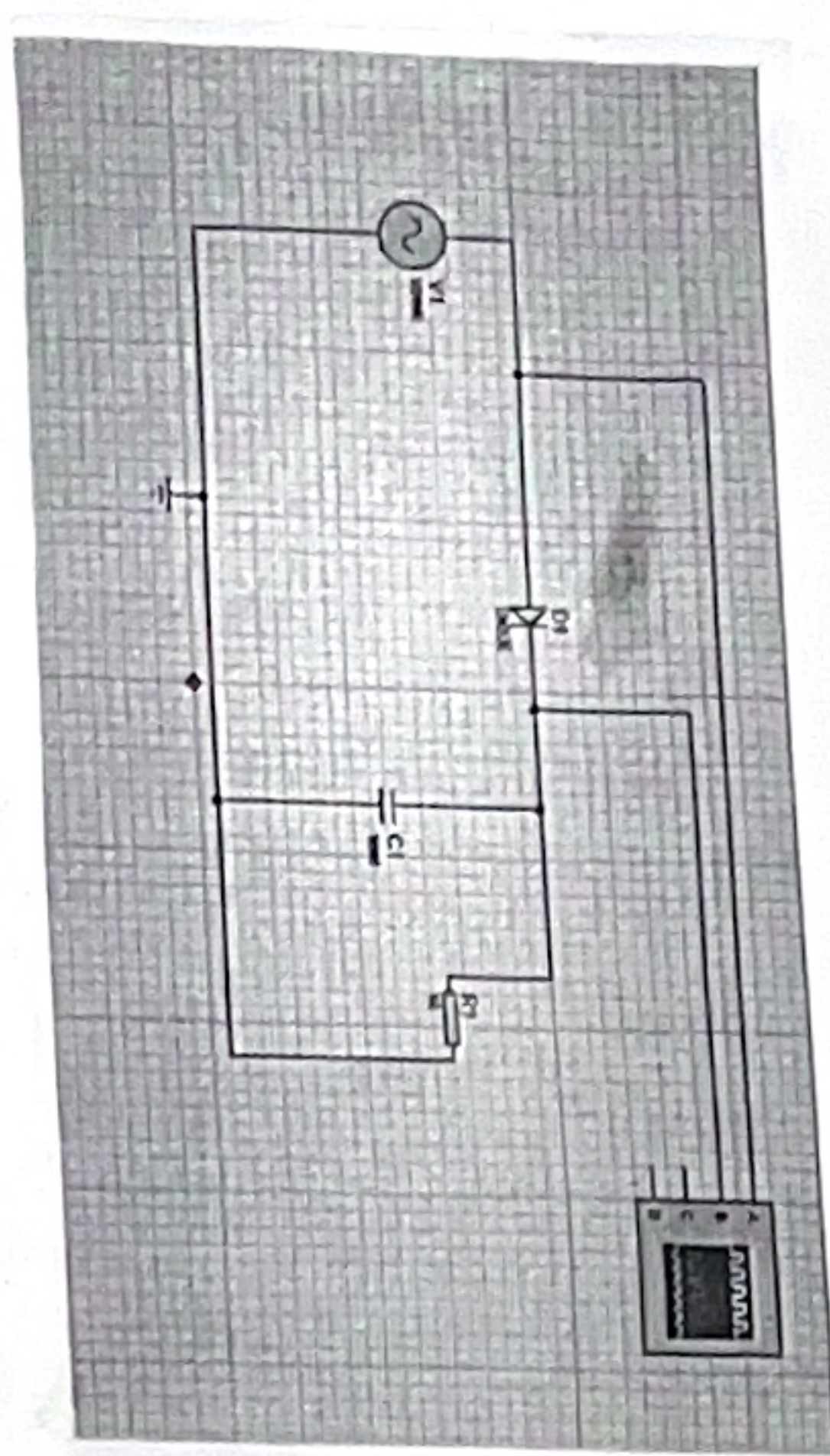
In order to get, pure D.C the output is filtered by a capacitor filter.

Procedure:

1. Drag the required components from the projects library.
2. Connect the components as per the circuit diagram.
3. Measure the input and output voltages and note in tabulations.
3. Draw the graph for input and output voltages.







Result:
Input and output waveforms of a half and full bridge-wave with or without filter are observed and plotted.